

GED Science Practice Test 15

PART A — QUESTIONS

GED-SCI15-001

Section: Science

Type: mcq

Category: Life Science

Skill: Designing and Interpreting Science Experiments

Prompt:

A researcher studies whether a daily vitamin D supplement affects calcium absorption. Two groups of adults eat the same diet for eight weeks. Group A takes a vitamin D supplement each day. Group B takes a capsule with no active ingredient. At the end of the study, calcium absorption is measured.

Group | Average calcium absorption (%)

A | 34

B | 24

Which conclusion is best supported by the data?

- A) The vitamin D group had higher average calcium absorption.
- B) The placebo group had higher average calcium absorption.
- C) Both groups absorbed the same amount of calcium.
- D) Calcium absorption cannot be measured in a study.

GED-SCI15-002

Section: Science

Type: mcq

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

Efficiency can be calculated as $\text{useful energy output} \div \text{total energy input} \times 100$.

Device | Total energy input (J) | Useful energy output (J)

A | 500 | 250

B | 400 | 260

C | 600 | 300

D | 800 | 320

Which device has the greatest efficiency?

- A) Device A
- B) Device B
- C) Device C
- D) Device D

GED-SCI15-003

Section: Science

Type: mcq

Category: Earth and Space Science

Skill: Reading for Meaning in Science

Prompt:

A lunar eclipse occurs when Earth moves between the Sun and the Moon. Earth's shadow falls on the Moon, causing the Moon to appear darker or reddish from Earth.

Which arrangement causes a lunar eclipse?

- A) Sun, Earth, Moon
- B) Sun, Moon, Earth
- C) Moon, Sun, Earth
- D) Earth, Moon, Sun

GED-SCI15-004

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

When body temperature rises above normal, sensors in the brain help trigger responses such as sweating and widening blood vessels near the skin. These responses help the body lose heat. As temperature returns toward normal, the responses decrease.

Which idea is best illustrated by this process?

- A) Negative feedback helps maintain body temperature.
- B) The body cannot respond to temperature changes.
- C) Sweating always increases body temperature.
- D) Blood vessels have no role in heat loss.

GED-SCI15-005

Section: Science

Type: numeric_entry

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A small electric motor uses 750 J of total energy and produces 300 J of useful mechanical energy.

Efficiency = useful energy output ÷ total energy input × 100

What is the motor's efficiency?

Type your answer as a percent rounded to the nearest whole number.

GED-SCI15-006

Section: Science

Type: multi_select

Category: Physical Science

Skill: Designing and Interpreting Science Experiments

Prompt:

Select TWO answers.

A student tests how the angle of a wind-turbine blade affects energy output. The same fan speed, turbine size, and measuring device are used for every trial. Only the blade angle changes.

Which TWO factors should be kept constant?

- A) Fan speed

- B) Turbine size
- C) Blade angle
- D) Energy output
- E) Final measured value

GED-SCI15-007

Section: Science

Type: dropdown

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

A trait appears only when a person inherits two changed copies of a gene, one from each parent. A person with one changed copy and one working copy does not show the trait but can pass the changed copy to children. This person is a [D1], and the trait is most likely [D2].

Dropdowns:

D1) carrier | producer | decomposer

D2) recessive | dominant | noninherited

GED-SCI15-008

Section: Science

Type: mcq

Category: Physical Science

Skill: Reading for Meaning in Science

Prompt:

A hydraulic lift uses liquid in connected cylinders. When force is applied to a small piston, pressure is transferred through the liquid to a larger piston. The larger piston can lift a heavy load because the same pressure acts over a greater area.

Which principle is mainly described?

- A) Pressure transmitted through a fluid can multiply force over a larger area.
- B) Liquids cannot transfer force.
- C) A larger piston removes gravity from an object.
- D) Hydraulic systems work only by changing DNA.

GED-SCI15-009

Section: Science

Type: mcq

Category: Earth and Space Science

Skill: Reading for Meaning in Science

Prompt:

A geothermal power plant uses heat from inside Earth. Hot underground rock heats water, and the resulting steam can turn a turbine to generate electricity. In some regions, this heat source can be used repeatedly if the system is managed carefully.

Which statement best describes geothermal energy?

- A) It uses internal heat from Earth to help generate electricity.
- B) It is made by burning fresh tree leaves.
- C) It comes only from reflected moonlight.
- D) It requires stopping all plate motion.

GED-SCI15-010

Section: Science

Type: mcq

Category: Life Science

Skill: Using Numbers and Graphics in Science

Prompt:

A clinic records parathyroid hormone level and blood calcium level in four patients.

Patient | Parathyroid hormone level | Blood calcium level

A | Normal | Normal

B | High | Low

C | Low | High

D | Normal | High

Parathyroid hormone can help raise blood calcium when calcium is low.

Which patient's results best match a response to low blood calcium?

A) Patient A

B) Patient B

C) Patient C

D) Patient D

GED-SCI15-011

Section: Science

Type: drag_drop

Category: Physical Science

Skill: Designing and Interpreting Science Experiments

Prompt:

Match each target with the correct tile.

Tiles:

A) Type of brake material used

B) Peak temperature reached by the brake pad

C) Same wheel speed before braking

D) Color of the brake handle

Targets:

T1) Independent variable

T2) Dependent variable

T3) Controlled variable

GED-SCI15-012

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

Apoptosis is a controlled process in which a cell breaks itself down without spilling harmful contents into nearby tissue. This process helps shape developing organs and removes damaged cells that could otherwise cause problems.

Which statement best describes apoptosis?

A) It is programmed cell death that can help remove damaged cells.

- B) It is the uncontrolled bursting of all cells in the body.
- C) It is the movement of magma through Earth's crust.
- D) It is the bending of light through a lens.

GED-SCI15-013

Section: Science

Type: mcq

Category: Physical Science

Skill: Reading for Meaning in Science

Prompt:

A ceramic mug and a metal pan are both placed in an oven. After heating, the metal pan becomes hot more quickly than the ceramic mug because metal transfers thermal energy faster through its particles.

Which property is mainly responsible for the difference?

- A) Thermal conductivity
- B) Chromosome number
- C) Ocean salinity
- D) Inherited variation

GED-SCI15-014

Section: Science

Type: numeric_entry

Category: Earth and Space Science

Skill: Using Numbers and Graphics in Science

Prompt:

A recycling center processes 7,500 kg of electronic waste in 5 days.

What is the average processing rate in kilograms per day?

Type your answer as a whole number.

GED-SCI15-015

Section: Science

Type: mcq

Category: Life Science

Skill: Designing and Interpreting Science Experiments

Prompt:

A researcher studies whether a new memory strategy improves recall. One group is taught the strategy, while a comparison group studies the same word list without the strategy. Both groups have the same study time and take the same recall test.

Why should both groups have the same study time?

- A) To help isolate the effect of the memory strategy
- B) To make the word list become shorter
- C) To guarantee both groups recall every word
- D) To prevent the use of a comparison group

GED-SCI15-016

Section: Science

Type: mcq

Category: Physical Science

Skill: Designing and Interpreting Science Experiments

Prompt:

A student tests how the mass of a cart affects its stopping time on a level track. The cart is released with the same starting speed in each trial. A braking pad applies the same force each time.

What is the independent variable?

- A) Mass of the cart
- B) Stopping time
- C) Starting speed
- D) Braking force

GED-SCI15-017

Section: Science

Type: dropdown

Category: Physical Science

Skill: Reading for Meaning in Science

Prompt:

A wheel and axle is a simple machine. Turning a larger wheel can help rotate a smaller axle with more turning effect. This can provide [D1], but the input often moves through a [D2] distance than the output.

Dropdowns:

D1) mechanical advantage | blood clotting | gene expression

D2) greater | zero-only | genetic

GED-SCI15-018

Section: Science

Type: mcq

Category: Life Science

Skill: Using Numbers and Graphics in Science

Prompt:

A researcher measures average blood glucose after two different lunches with the same calories.

Lunch type | Average blood glucose 1 hour after lunch (mg/dL)

High refined carbohydrate | 156

High fiber and protein | 122

Which statement is best supported by the data?

- A) The high fiber and protein lunch was associated with lower blood glucose after 1 hour.
- B) The high refined carbohydrate lunch was associated with lower blood glucose after 1 hour.
- C) Both lunches produced the same average blood glucose.
- D) Blood glucose cannot change after a meal.

GED-SCI15-019

Section: Science

Type: mcq

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A student compares how far identical toy cars travel after being launched with different amounts of stored elastic energy.

Stored elastic energy (J): 1, 3, 5, 7
Travel distance (m): 0.8, 2.2, 3.7, 5.1

Which statement is best supported by the data?

- A) Greater stored elastic energy was associated with greater travel distance.
- B) Greater stored elastic energy was associated with shorter travel distance.
- C) Travel distance stayed exactly the same for all energy values.
- D) Stored elastic energy had no relationship to motion.

GED-SCI15-020

Section: Science

Type: multi_select

Category: Earth and Space Science

Skill: Using Numbers and Graphics in Science

Prompt:

Select TWO answers.

A region tracks two climate-related measurements.

Decade	Average growing season length (days)	Average annual frost days
1980s	154	92
1990s	161	84
2000s	169	76
2010s	177	68

Which TWO statements are best supported by the data?

- A) Average growing season length increased over the decades.
- B) Average annual frost days decreased over the decades.
- C) Frost days increased as growing season length increased.
- D) The 1980s had the longest growing season.
- E) Both measurements stayed exactly the same.

GED-SCI15-021

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

Stem cells are cells that can divide and, under certain conditions, develop into specialized cell types. For example, some stem cells in bone marrow can develop into different kinds of blood cells.

Which statement best describes stem cells?

- A) They can give rise to specialized cells.
- B) They are minerals found only in rocks.
- C) They are gases released by volcanoes.
- D) They prevent all tissue repair.

GED-SCI15-022

Section: Science

Type: mcq

Category: Physical Science

Skill: Reading for Meaning in Science

Prompt:

A student compares two balls dropped from the same height onto different mats. One mat brings the ball to a stop quickly, while another mat increases the stopping time. Increasing stopping time can reduce the average impact force.

Which statement best explains why padding can reduce injury risk?

- A) Padding increases stopping time, which can reduce impact force.
- B) Padding removes the object's mass.
- C) Padding prevents energy from being transferred in any way.
- D) Padding makes gravity act upward only.

GED-SCI15-023

Section: Science

Type: mcq

Category: Earth and Space Science

Skill: Reading for Meaning in Science

Prompt:

A geologist studies a hotspot volcanic island chain. The hotspot stays relatively fixed while an oceanic plate moves slowly over it. Older islands are farther from the active volcano, while the youngest island is above or near the hotspot.

Which statement is best supported by this model?

- A) Plate motion can create a chain of volcanic islands over a hotspot.
- B) Hotspots form only when rivers deposit sediment.
- C) All islands in a hotspot chain must be the same age.
- D) Plate motion stops when magma reaches the surface.

GED-SCI15-024

Section: Science

Type: numeric_entry

Category: Life Science

Skill: Using Numbers and Graphics in Science

Prompt:

A medical survey includes 640 adults. Of these adults, 96 report a family history of autoimmune disease.

What percent of the adults report a family history of autoimmune disease?

Type your answer as a percent rounded to the nearest whole number.

GED-SCI15-025

Section: Science

Type: mcq

Category: Physical Science

Skill: Designing and Interpreting Science Experiments

Prompt:

A student tests how the surface area of a parachute affects landing speed. The same mass is attached to parachutes with different surface areas. Each parachute is dropped from the same height, and landing speed is measured.

Which factor is the dependent variable?

- A) Landing speed
- B) Parachute surface area
- C) Same attached mass
- D) Same drop height

GED-SCI15-026

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

The pancreas has cells that release insulin into the blood. Insulin helps many body cells take in glucose. If insulin production is too low or cells do not respond well to insulin, blood glucose may remain too high.

Which statement best describes insulin's role?

- A) It helps cells take in glucose from the blood.
- B) It carries oxygen in red blood cells.
- C) It produces sound waves in the lungs.
- D) It changes chromosomes into minerals.

GED-SCI15-027

Section: Science

Type: mcq

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A student records the distance traveled by a sliding block after different amounts of friction-reducing wax are applied to the same surface.

Wax applied (layers): 0, 1, 2, 3

Sliding distance (m): 1.2, 1.9, 2.7, 3.4

Which statement is best supported by the data?

- A) More wax layers were associated with a longer sliding distance.
- B) More wax layers were associated with a shorter sliding distance.
- C) The block traveled exactly the same distance every time.
- D) Wax made the block unable to move.

GED-SCI15-028

Section: Science

Type: mcq

Category: Earth and Space Science

Skill: Using Numbers and Graphics in Science

Prompt:

A student compares the orbital periods of four objects orbiting the Sun.

Object | Orbital period (Earth years)

A | 0.6

B | 1.0

C | 11.9
D | 84.0

Which object has the longest orbital period?

- A) Object A
- B) Object B
- C) Object C
- D) Object D

GED-SCI15-029

Section: Science

Type: mcq

Category: Life Science

Skill: Using Numbers and Graphics in Science

Prompt:

A doctor records average liver enzyme levels before and after patients stop taking a medication that was stressing the liver.

Time | Average liver enzyme level (units/L)

Before stopping medication | 96

2 weeks after | 62

4 weeks after | 38

Which statement is best supported by the data?

- A) Average liver enzyme level decreased after the medication was stopped.
- B) Average liver enzyme level increased after the medication was stopped.
- C) Liver enzyme level stayed exactly the same.
- D) Liver enzymes disappeared immediately.

GED-SCI15-030

Section: Science

Type: mcq

Category: Physical Science

Skill: Reading for Meaning in Science

Prompt:

A student separates a mixture of gravel and fine sand using a screen. The larger gravel pieces stay on the screen, while the smaller sand grains pass through.

Which property is mainly used for separation?

- A) Particle size
- B) Gene sequence
- C) Blood pressure
- D) Orbital period

GED-SCI15-031

Section: Science

Type: multi_select

Category: Life Science

Skill: Designing and Interpreting Science Experiments

Prompt:

Select TWO answers.

A researcher studies whether daily step count affects resting heart rate. Volunteers wear step counters for four weeks, and resting heart rate is measured each morning.

Which TWO steps would improve the investigation?

- A) Use the same type of step counter for all volunteers.
- B) Measure resting heart rate at the same time of day.
- C) Change the heart-rate method for every volunteer.
- D) Record only the volunteers with the lowest heart rate.
- E) Allow some volunteers to skip wearing the counter.

GED-SCI15-032

Section: Science

Type: mcq

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A student compares the frequency of waves with different periods.

Wave | Period (s)

A | 0.2

B | 0.5

C | 1.0

D | 2.0

Frequency = $1 \div \text{period}$

Which wave has a frequency of 5 Hz?

- A) Wave A
- B) Wave B
- C) Wave C
- D) Wave D

GED-SCI15-033

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

A coral reef is damaged by a severe storm, but living coral fragments and reef structure remain. Over time, algae, young corals, and other organisms recolonize open spaces on the reef.

Which ecological process is most directly described?

- A) Secondary succession
- B) Nuclear fusion
- C) Mechanical advantage
- D) Refraction

GED-SCI15-034

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

A person inherits a changed copy of a gene involved in making a receptor protein on cell surfaces. The altered receptor binds its signal molecule poorly, so the cell responds less strongly to that signal.

Which statement best explains the effect?

- A) A gene change can alter a receptor protein and affect cell signaling.
- B) Receptor proteins are not affected by genes.
- C) Cell signaling happens only in rocks.
- D) A changed gene always improves receptor function.

PART B — ANSWER KEY + EXPLANATIONS

GED-SCI15-001 — Correct: A — Correct Answer: The vitamin D group had higher average calcium absorption. — Explanation: Group A took vitamin D and had higher average calcium absorption than Group B.

GED-SCI15-002 — Correct: B — Correct Answer: Device B — Explanation: Efficiency values are A = 50%, B = 65%, C = 50%, and D = 40%, so Device B has the greatest efficiency.

GED-SCI15-003 — Correct: A — Correct Answer: Sun, Earth, Moon — Explanation: A lunar eclipse occurs when Earth is between the Sun and the Moon.

GED-SCI15-004 — Correct: A — Correct Answer: Negative feedback helps maintain body temperature. — Explanation: Responses such as sweating decrease as temperature returns toward normal, showing negative feedback.

GED-SCI15-005 — Correct: 40 — Tolerance: exact — Correct Answer: 40 — Explanation: $\text{Efficiency} = 300 \div 750 \times 100 = 40\%$.

GED-SCI15-006 — Correct: A,B — Correct Answer: Fan speed | Turbine size — Explanation: These factors should stay constant so blade angle is the main variable being tested.

GED-SCI15-007 — Correct: D1=carrier; D2=recessive — Correct Answer: D1=carrier; D2=recessive — Explanation: A person with one changed copy can carry a recessive allele without showing the trait.

GED-SCI15-008 — Correct: A — Correct Answer: Pressure transmitted through a fluid can multiply force over a larger area. — Explanation: Hydraulic systems transfer pressure through fluid, allowing a larger piston to produce a larger force.

GED-SCI15-009 — Correct: A — Correct Answer: It uses internal heat from Earth to help generate electricity. — Explanation: Geothermal systems use Earth's internal heat to produce steam and generate electricity.

GED-SCI15-010 — Correct: B — Correct Answer: Patient B — Explanation: Patient B has low calcium and high parathyroid hormone, matching the response described.

GED-SCI15-011 — Correct: T1=A; T2=B; T3=C — Correct Answer: T1=Type of brake material used; T2=Peak temperature reached by the brake pad; T3=Same wheel speed before braking —

Explanation: Brake material is changed, peak temperature is measured, and starting wheel speed is controlled.

GED-SCI15-012 — Correct: A — Correct Answer: It is programmed cell death that can help remove damaged cells. — Explanation: Apoptosis is controlled cell death that helps remove damaged or unnecessary cells.

GED-SCI15-013 — Correct: A — Correct Answer: Thermal conductivity — Explanation: Metal conducts thermal energy faster than ceramic, so it heats more quickly.

GED-SCI15-014 — Correct: 1500 — Tolerance: exact — Correct Answer: 1500 — Explanation: $7,500 \text{ kg} \div 5 \text{ days} = 1,500 \text{ kg/day}$.

GED-SCI15-015 — Correct: A — Correct Answer: To help isolate the effect of the memory strategy — Explanation: Keeping study time the same helps test the strategy rather than time spent studying.

GED-SCI15-016 — Correct: A — Correct Answer: Mass of the cart — Explanation: The independent variable is the factor intentionally changed by the student.

GED-SCI15-017 — Correct: D1=mechanical advantage; D2=greater — Correct Answer: D1=mechanical advantage; D2=greater — Explanation: A wheel and axle can provide mechanical advantage, often requiring the input to move through a greater distance.

GED-SCI15-018 — Correct: A — Correct Answer: The high fiber and protein lunch was associated with lower blood glucose after 1 hour. — Explanation: The high fiber and protein lunch had a lower average blood glucose value, 122 mg/dL, than the high refined carbohydrate lunch.

GED-SCI15-019 — Correct: A — Correct Answer: Greater stored elastic energy was associated with greater travel distance. — Explanation: Travel distance increased as stored elastic energy increased.

GED-SCI15-020 — Correct: A,B — Correct Answer: Average growing season length increased over the decades. | Average annual frost days decreased over the decades. — Explanation: Growing season length increased from 154 to 177 days, while frost days decreased from 92 to 68.

GED-SCI15-021 — Correct: A — Correct Answer: They can give rise to specialized cells. — Explanation: Stem cells can divide and develop into specialized cell types.

GED-SCI15-022 — Correct: A — Correct Answer: Padding increases stopping time, which can reduce impact force. — Explanation: Increasing the time over which an object stops can lower the average force during impact.

GED-SCI15-023 — Correct: A — Correct Answer: Plate motion can create a chain of volcanic islands over a hotspot. — Explanation: As a plate moves over a relatively fixed hotspot, volcanic islands of different ages can form in a chain.

GED-SCI15-024 — Correct: 15 — Tolerance: exact — Correct Answer: 15 — Explanation: $96 \div 640 \times 100 = 15\%$.

GED-SCI15-025 — Correct: A — Correct Answer: Landing speed — Explanation: The dependent variable is the measured result after changing parachute surface area.

GED-SCI15-026 — Correct: A — Correct Answer: It helps cells take in glucose from the blood. — Explanation: Insulin helps body cells take up glucose, lowering blood glucose.

GED-SCI15-027 — Correct: A — Correct Answer: More wax layers were associated with a longer sliding distance. — Explanation: The sliding distance increased as more wax layers were applied.

GED-SCI15-028 — Correct: D — Correct Answer: Object D — Explanation: Object D has the longest listed orbital period, 84.0 Earth years.

GED-SCI15-029 — Correct: A — Correct Answer: Average liver enzyme level decreased after the medication was stopped. — Explanation: The average liver enzyme level fell from 96 to 38 units/L over four weeks.

GED-SCI15-030 — Correct: A — Correct Answer: Particle size — Explanation: Screening separates particles based on size.

GED-SCI15-031 — Correct: A,B — Correct Answer: Use the same type of step counter for all volunteers. | Measure resting heart rate at the same time of day. — Explanation: Consistent equipment and measurement timing improve reliability.

GED-SCI15-032 — Correct: A — Correct Answer: Wave A — Explanation: $\text{Frequency} = 1 \div 0.2 \text{ s} = 5 \text{ Hz}$.

GED-SCI15-033 — Correct: A — Correct Answer: Secondary succession — Explanation: Secondary succession occurs after disturbance when living material or structure remains.

GED-SCI15-034 — Correct: A — Correct Answer: A gene change can alter a receptor protein and affect cell signaling. — Explanation: Genes provide instructions for proteins, so a gene change can alter receptor function and cell signaling.